

Dimension reduction for genomics

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Dependencies

This document depends on the following packages:

```
library(devtools)
library(Biobase)
```

To install these packages you can use the code (or if you are compiling the document, remove the `eval=FALSE` from the chunk.)

```
install.packages(c("devtools"))
source("http://www.bioconductor.org/biocLite.R")
biocLite(c("Biobase"))
```

General principles

- Can we find patterns in matrices of data?

Load some data

We will use this expression set that combines two studies Transcriptome genetics using second generation sequencing in a Caucasian population. (<http://www.ncbi.nlm.nih.gov/pubmed?term=20220756%5Buid%5D>) and Understanding mechanisms underlying human gene expression variation with RNA sequencing. (<http://www.ncbi.nlm.nih.gov/pubmed?term=20220758>). These studies are different populations but we counted the same genes for both. Then we'll explore the differences.

```

con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/montpick_eset.RData")
load(file=con)
close(con)
mp = montpick.eset
pdata=pData(mp)
edata=as.data.frame(exprs(mp))
fdata = fData(mp)
ls()

```

```

## [1] "con"          "edata"        "fdata"        "montpick.eset"
## [5] "mp"          "pdata"        "tropical"

```

Calculate the singular vectors

Here we calculate the singular vectors:

```

edata = edata[rowMeans(edata) > 100, ]
edata = log2(edata + 1)
edata_centered = edata - rowMeans(edata)
svd1 = svd(edata_centered)
names(svd1)

```

```

## [1] "d" "u" "v"

```

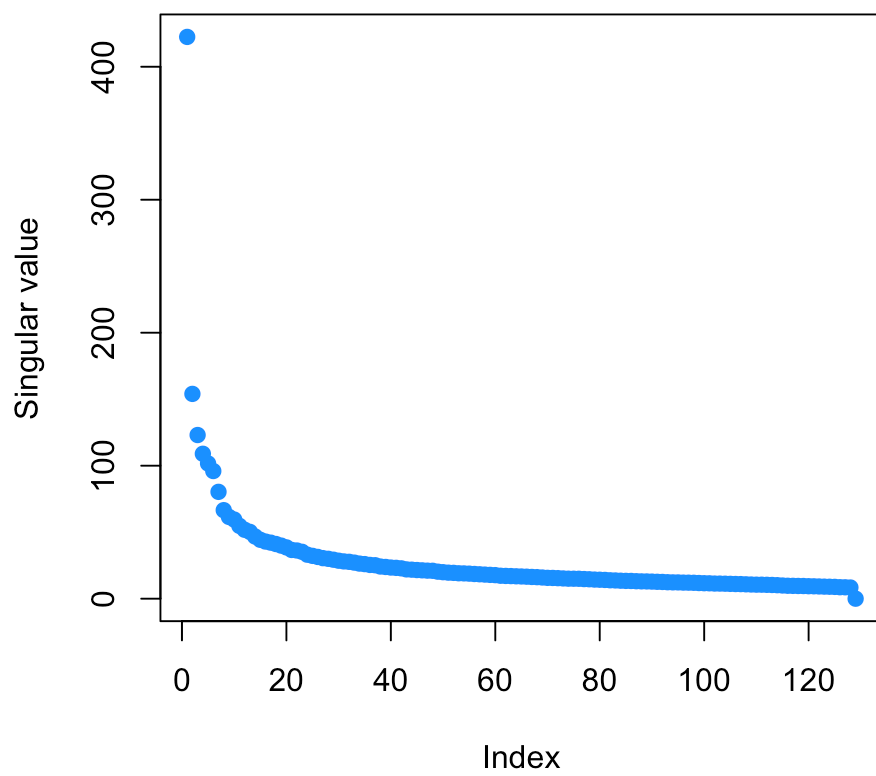
Look at the percent variance explained

The percent of variance explained is given by $\frac{d_{ii}}{\sum_j d_{jj}^2}$

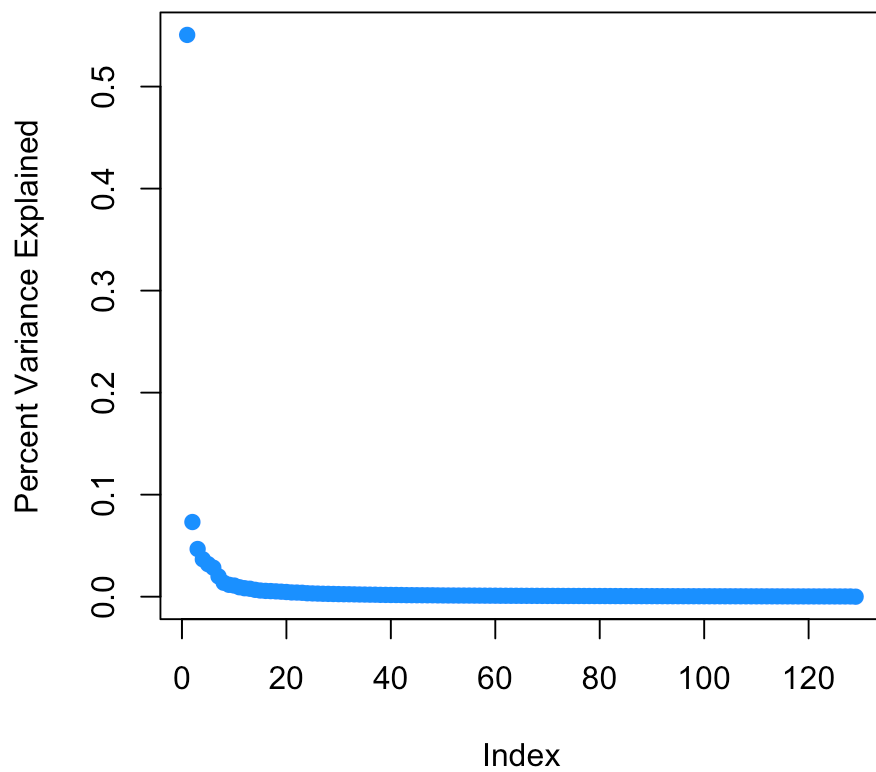
```

plot(svd1$d,ylab="Singular value",col=2)

```

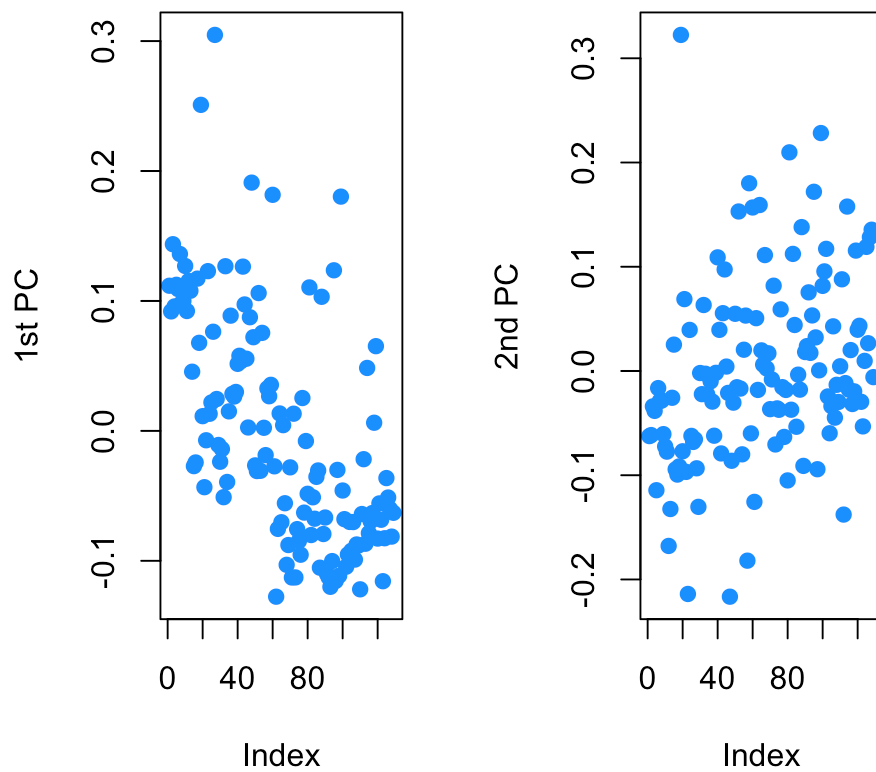


```
plot(svd1$d^2/sum(svd1$d^2),ylab="Percent Variance Explained",col=2)
```



Plot top two principal components

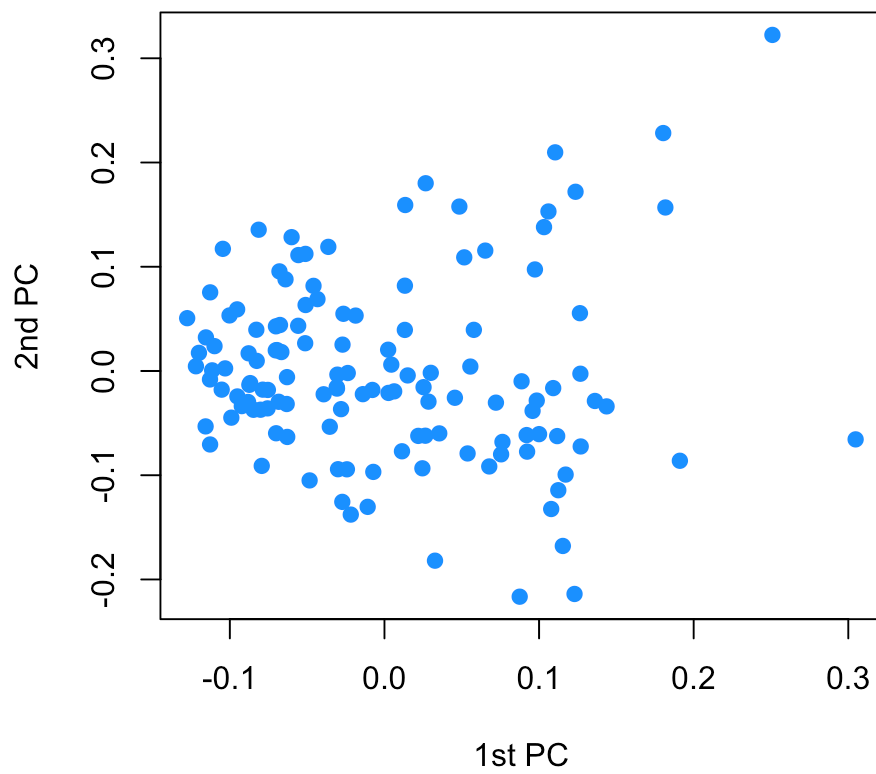
```
par(mfrow=c(1,2))  
plot(svd1$v[,1],col=2,ylab="1st PC")  
plot(svd1$v[,2],col=2,ylab="2nd PC")
```



Plot PC1 vs. PC2

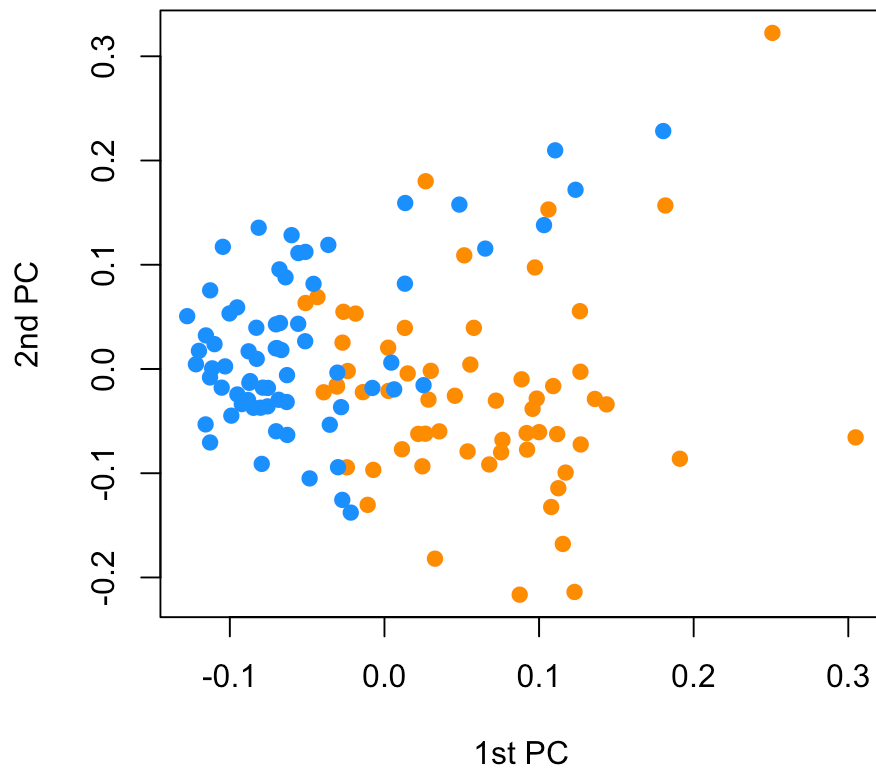
A very common plot is to plot PC1 versus PC2 to see if you can see any “clusters” or “groups”.

```
plot(svd1$v[,1],svd1$v[,2],col=2,ylab="2nd PC",xlab="1st PC")
```



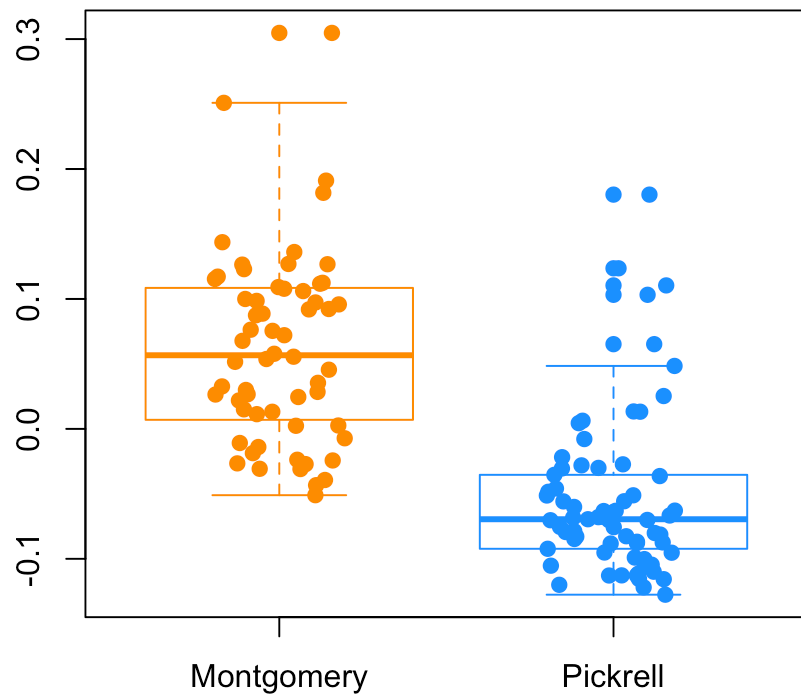
One thing you can do is color them by different variables to see if clusters stand out.

```
plot(svd1$v[,1],svd1$v[,2],ylab="2nd PC",  
     xlab="1st PC",col=as.numeric(pdata$study))
```



Another common plot is to make boxplots comparing the PC for different levels of known covariates (don't forget to show the actual data!).

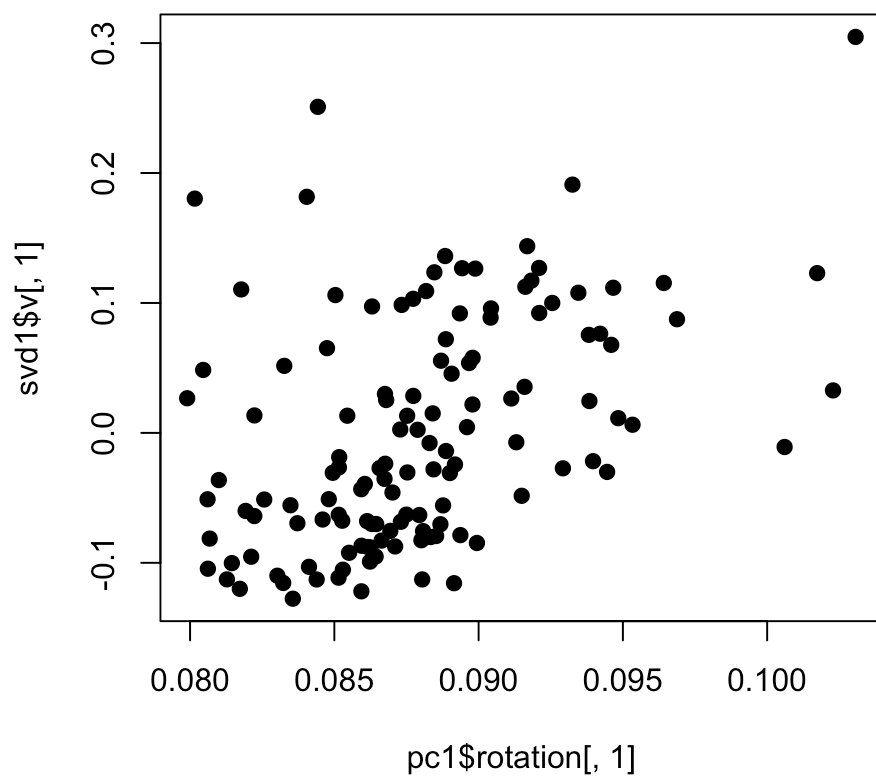
```
boxplot(svd1$v[,1] ~ pdata$study, border=c(1,2))  
points(svd1$v[,1] ~ jitter(as.numeric(pdata$study)), col=as.numeric(pdata$study))
```



PCs versus SVs

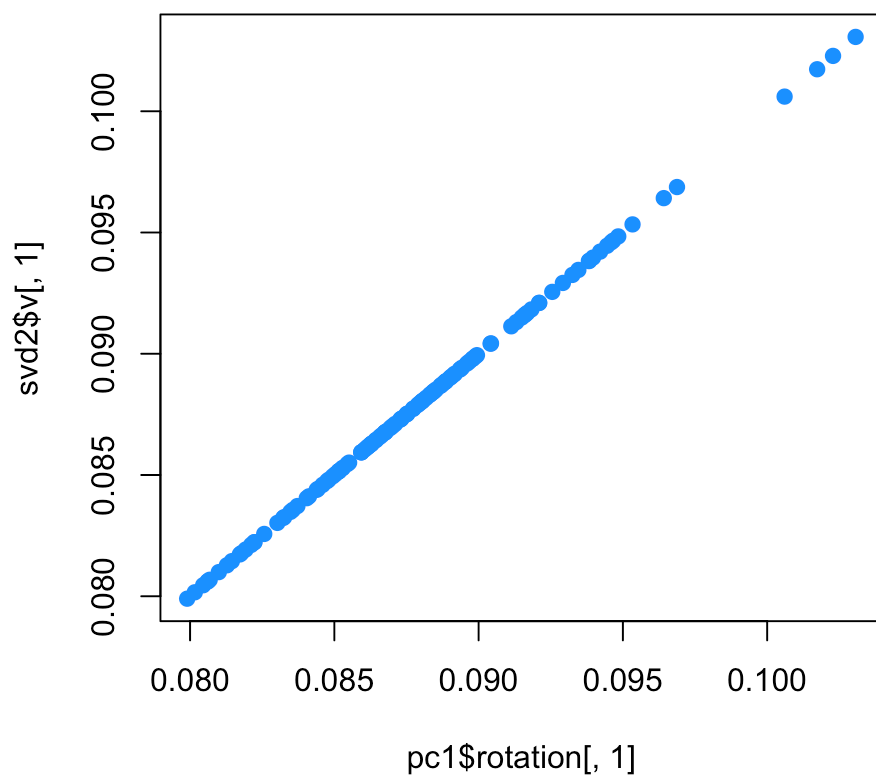
What we have been plotting is not exactly the principal components.

```
pc1 = prcomp(edata)
plot(pc1$rotation[,1],svd1$v[,1])
```

To get the actual PCs you have to subtract the column means rather than the row means when normalizing.

```
edata_centered2 = t(t(edata) - colMeans(edata))
svd2 = svd(edata_centered2)
plot(pc1$rotation[,1],svd2$sv[,1],col=2)
```



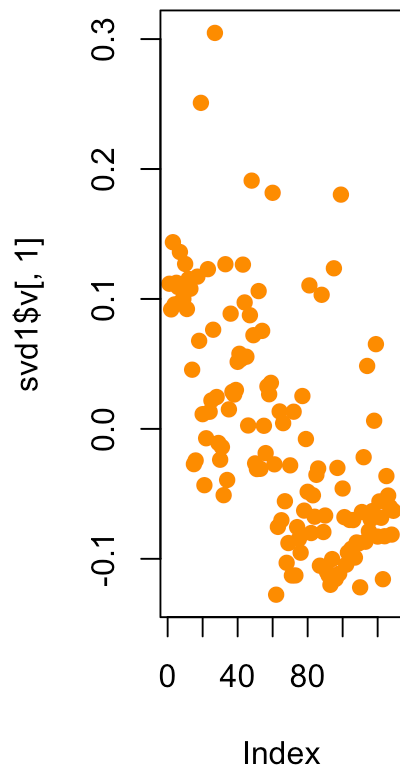
Despite this, it is most common for people to perform row-centering and then plot the singular vectors (sometimes labeling them PCs like I have done in this document)

Outliers

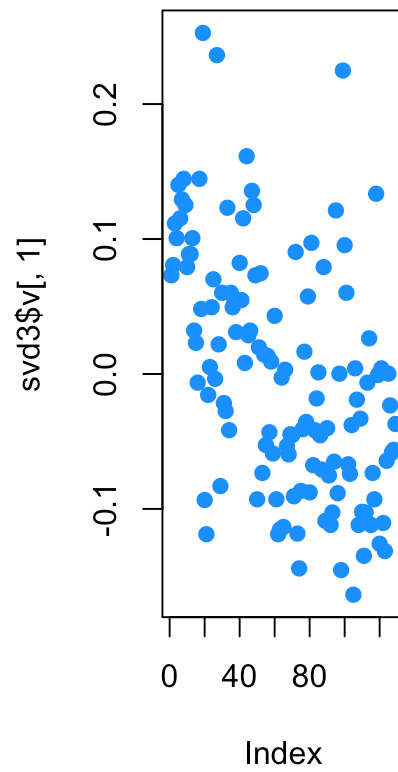
What happens if we introduce a single outlying gene

```
edat_outlier = edat_centered
edat_outlier[1,] = edat_centered[1,] * 10000
svd3 = svd(edat_outlier)
par(mfrow=c(1,2))
plot(svd1$sv[,1],col=1,main="Without outlier")
plot(svd3$sv[,1],col=2,main="With outlier")
```

Without outlier

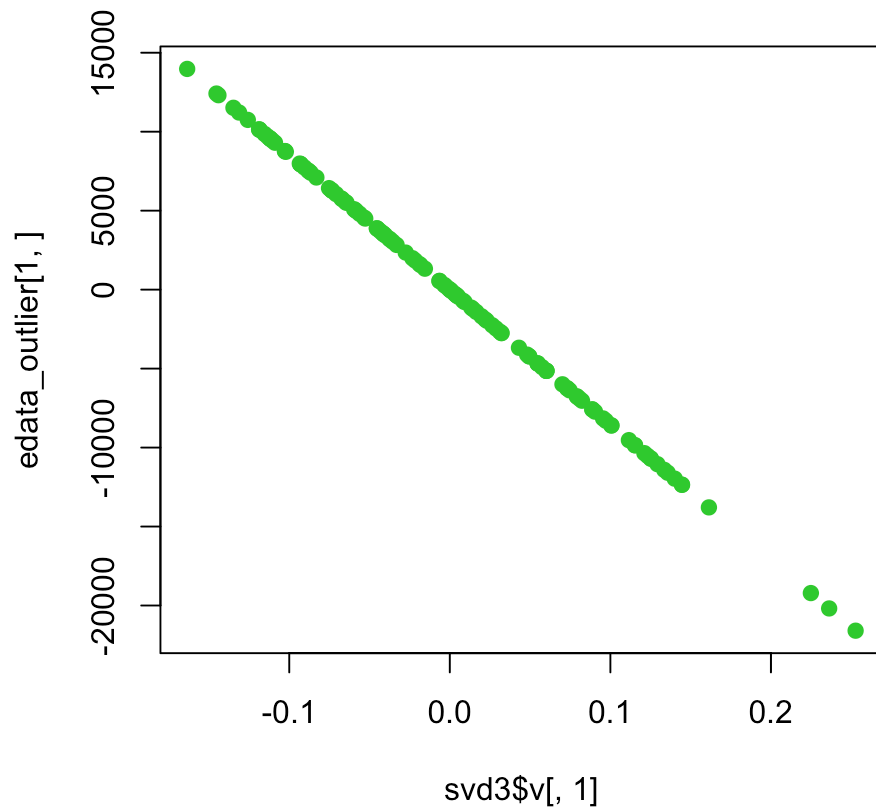


With outlier



It turns out the new top singular vector is perfectly correlated with the outlying gene

```
plot(svd3$v[,1],edata_outlier[1,],col=4)
```



Further resources

There are a large number of resources available about PCA and SVD but the lecture notes from Advanced Statistics for the Life Sciences (<http://genomicsclass.github.io/book/>) are the best set of lecture notes focused on genomics currently available.

Session information

Here is the session information

```
devtools::session_info()
```

```

## setting value
## version R version 3.2.1 (2015-06-18)
## system x86_64, darwin10.8.0
## ui RStudio (0.99.447)
## language (EN)
## collate en_US.UTF-8
## tz America/New_York
##
## package * version date
## acepack 1.3-3.3 2014-11-24
## annotate 1.46.1 2015-07-11
## AnnotationDbi * 1.30.1 2015-04-26
## assertthat 0.1 2013-12-06
## BiasedUrn * 1.06.1 2013-12-29
## Biobase * 2.28.0 2015-04-17
## BiocGenerics * 0.14.0 2015-04-17
## BiocInstaller * 1.18.4 2015-07-22
## BiocParallel 1.2.20 2015-08-07
## biomaRt 2.24.0 2015-04-17
## Biostrings 2.36.3 2015-08-12
## bitops 1.0-6 2013-08-17
## bladderbatch * 1.6.0 2015-08-26
## broom * 0.3.7 2015-05-06
## caTools 1.17.1 2014-09-10
## cluster 2.0.3 2015-07-21
## colorspace 1.2-6 2015-03-11
## corpcor 1.6.8 2015-07-08
## curl 0.9.2 2015-08-08
## DBI * 0.3.1 2014-09-24
## dendextend * 1.1.0 2015-07-31
## DESeq2 * 1.8.1 2015-05-02
## devtools * 1.8.0 2015-05-09
## digest 0.6.8 2014-12-31
## dplyr * 0.4.3 2015-09-01
## edge * 2.1.0 2015-09-06
## evaluate 0.7.2 2015-08-13
## foreign 0.8-66 2015-08-19
## formatR 1.2 2015-04-21
## Formula * 1.2-1 2015-04-07
## futile.logger 1.4.1 2015-04-20
## futile.options 1.0.0 2010-04-06
## gdata 2.17.0 2015-07-04
## genefilter * 1.50.0 2015-04-17
## geneLenDataBase * 1.4.0 2015-09-06
## geneplotter 1.46.0 2015-04-17
## GenomeInfoDb * 1.4.2 2015-08-15
## GenomicAlignments 1.4.1 2015-04-24
## GenomicFeatures 1.20.2 2015-08-14
## GenomicRanges * 1.20.5 2015-06-09
## genstats * 0.1.02 2015-09-05
## ggplot2 * 1.0.1 2015-03-17
## git2r 0.11.0 2015-08-12

```

##	GO.db	3.1.2	2015-09-06
##	goseq	* 1.20.0	2015-04-17
##	gplots	* 2.17.0	2015-05-02
##	gridExtra	2.0.0	2015-07-14
##	gtable	0.1.2	2012-12-05
##	gtools	3.5.0	2015-05-29
##	highr	0.5	2015-04-21
##	HistData	* 0.7-5	2014-04-26
##	Hmisc	* 3.16-0	2015-04-30
##	htmltools	0.2.6	2014-09-08
##	httr	1.0.0	2015-06-25
##	IRanges	* 2.2.7	2015-08-09
##	KernSmooth	2.23-15	2015-06-29
##	knitr	* 1.11	2015-08-14
##	lambda.r	1.1.7	2015-03-20
##	lattice	* 0.20-33	2015-07-14
##	latticeExtra	0.6-26	2013-08-15
##	lazyeval	0.1.10	2015-01-02
##	limma	* 3.24.15	2015-08-06
##	lme4	1.1-9	2015-08-20
##	locfit	1.5-9.1	2013-04-20
##	magrittr	1.5	2014-11-22
##	MASS	* 7.3-43	2015-07-16
##	Matrix	* 1.2-2	2015-07-08
##	MatrixEQTL	* 2.1.1	2015-02-03
##	memoise	0.2.1	2014-04-22
##	mgcv	* 1.8-7	2015-07-23
##	minqa	1.2.4	2014-10-09
##	mnormt	1.5-3	2015-05-25
##	munsell	0.4.2	2013-07-11
##	nlme	* 3.1-122	2015-08-19
##	nloptr	1.0.4	2014-08-04
##	nnet	7.3-10	2015-06-29
##	org.Hs.eg.db	* 3.1.2	2015-07-17
##	plyr	1.8.3	2015-06-12
##	preprocessCore	* 1.30.0	2015-04-17
##	proto	0.3-10	2012-12-22
##	psych	1.5.6	2015-07-08
##	qvalue	* 2.0.0	2015-04-17
##	R6	2.1.1	2015-08-19
##	RColorBrewer	1.1-2	2014-12-07
##	Rcpp	* 0.12.0	2015-07-25
##	RcppArmadillo	* 0.5.400.2.0	2015-08-17
##	RCurl	1.95-4.7	2015-06-30
##	reshape2	1.4.1	2014-12-06
##	rmarkdown	0.7	2015-06-13
##	rpart	4.1-10	2015-06-29
##	Rsamtools	1.20.4	2015-06-01
##	RSkittleBrewer	* 1.1	2015-09-05
##	RSQLite	* 1.0.0	2014-10-25
##	rstudioapi	0.3.1	2015-04-07
##	rtracklayer	1.28.9	2015-08-19

##	rversions	1.0.2	2015-07-13
##	S4Vectors	* 0.6.5	2015-09-01
##	scales	0.3.0	2015-08-25
##	snm	1.16.0	2015-04-17
##	snpStats	* 1.18.0	2015-04-17
##	stringi	0.5-5	2015-06-29
##	stringr	1.0.0	2015-04-30
##	survival	* 2.38-3	2015-07-02
##	sva	* 3.14.0	2015-04-17
##	tidyr	0.2.0	2014-12-05
##	UsingR	* 2.0-5	2015-08-06
##	whisker	0.3-2	2013-04-28
##	XML	3.98-1.3	2015-06-30
##	xml2	0.1.2	2015-09-01
##	xtable	1.7-4	2014-09-12
##	XVector	0.8.0	2015-04-17
##	yaml	2.1.13	2014-06-12
##	zlibbioc	1.14.0	2015-04-17
##	source		
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## Bioconductor
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It is also useful to compile the time the document was processed. This document was processed on: 2015-09-06.